

ASPIRE · TECH/BIO workflow

YAML configuration · Nextflow / Mamba · resumable execution · audited outputs

Read quality control

FASTP_QC → MERGE_READS → FILTER_READS → RELABEL_FILTERED

All biological samples and controls share read processing.



ASV construction

CONCAT_FASTAS → DEREPLICATE → DENOISE → CHIMERA_CHECK → CREATE_COUNT_MATRIX

Preserve original ASV counts and sequences, including control-only ASVs.



Alignment and full taxonomy

SINA_TRIM → TAXONOMY

Assign taxonomy to all inferred ASVs before biological feature filtering.



CONTROL_DECONTAM · optional, metadata-defined

Biological samples: ≥5,000 post-QC reads; nonzero controls: no depth cutoff

TECH: biological + blanks. BIO: same biological cohort + sample controls.

Independent prevalence scores < configured cutoffs → union ASV removal.

Retained counts unchanged. Positive controls: QC only. Unused arms can be disabled.



Reference screening

PREPARE_BLAST_DATABASES → MITOMASTER → MITO_DECONTAM

Local host / mitochondrial evidence; external MITOMASTER lookup is optional.



FILTER_ASVS · combined biological feature filtering

Initial abundance / nonzero → group and non-target → abundance / taxonomy → final nonzero

Mock: ≥0.1% RA in any biological sample AND nonzero counts in ≥5% of biological samples.

No second depth cutoff after CONTROL_DECONTAM. Final counts and FASTA agree.



PLOT_METADATA · final biological tables

ASV_target.tsv → synchronized long / wide tables and metadata

CAMI: Airways and Oral only; Skin BIO controls and TECH blanks excluded.



Optional analysis-table preparation

Group diagnostics / validated labels → batch correction and count selection

Downstream analyses use the selected counts and synchronized annotations.



Optional analysis branches · dependencies govern execution

Diversity / ordination · indicators · patient / paired contrasts · group-effect power

ASV–VOC associations · SPIEC-EASI → modules / topology → annotated networks

Final filtered FASTA + genome references → ASV–MAG links and network overlays.



Publish results and provenance

Module tables / plots → master summaries → HTML report and checksum inventories

GENERAL_STATS runs as a parallel QC branch; audit intermediates remain available.

Arrows show the main data path; enabled analyses may run concurrently. Control decontamination can be bypassed.